CO3043 - Mobile App Development

CHAPTER 1

Introduction to Mobile Development

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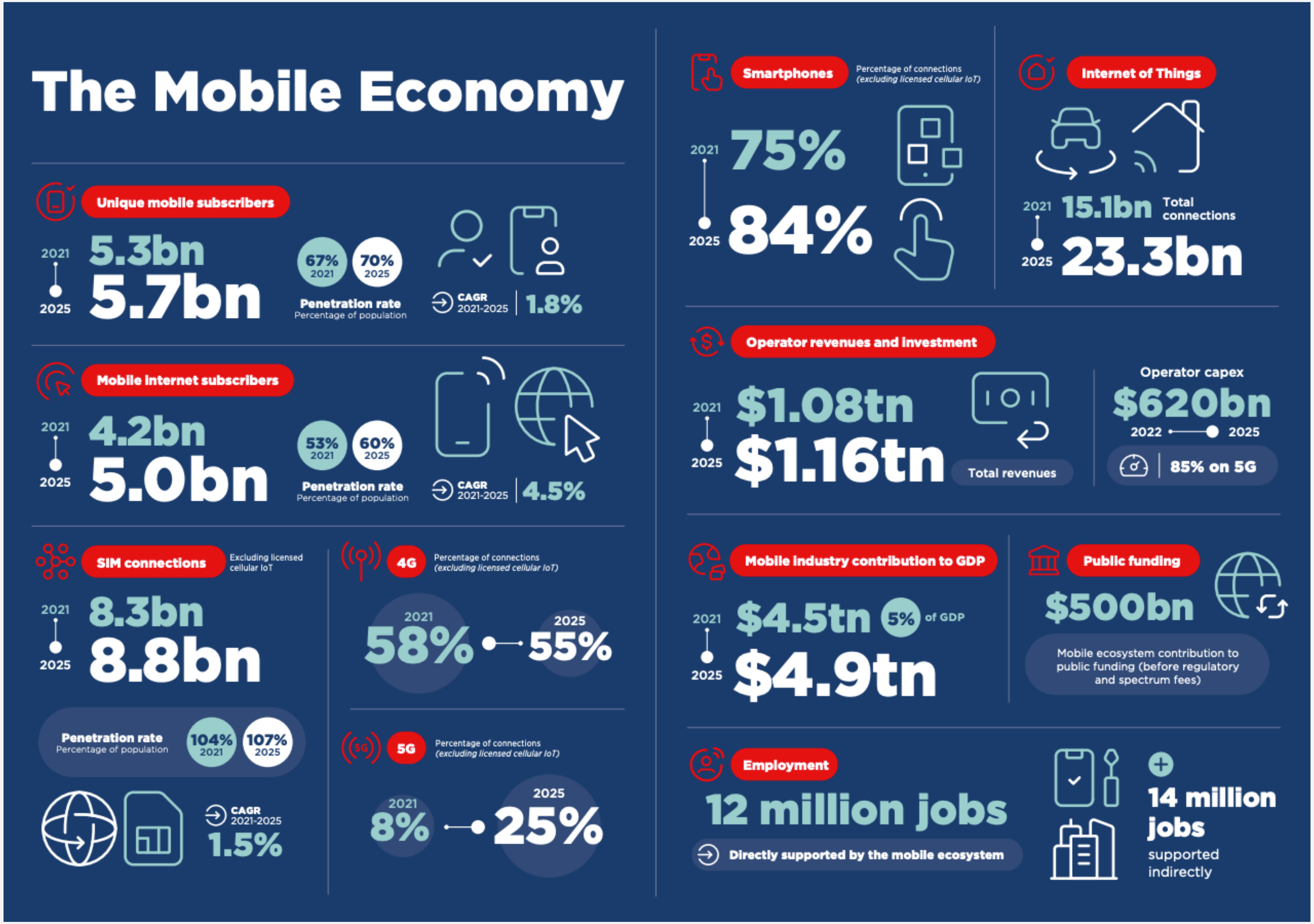
Contents

1.Why Mobile App?

The smartphone industry has been steadily developing and growing since 2008, both in market size and in number of models and vendors.

The global mobile application market size amounted to **USD 187.58 billion** in 2021 and is projected to grow at a compound annual growth rate of 13.4% from 2022 to 2030.



Source: GSMA Mobile Economy 2022

APPOTA
Entertainment Ecosystem

BÁO CÁO ỨNG DỤNG
DI ĐỘNG 2021

70%
Dân số sử dụng điện thoại di động

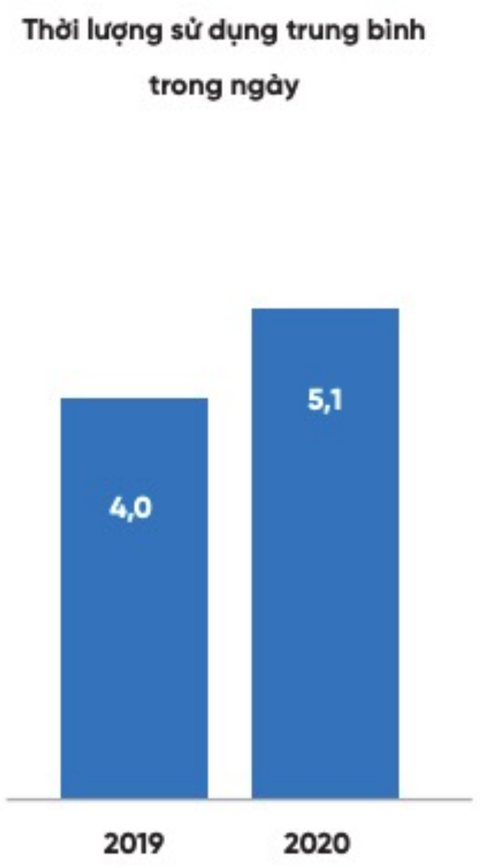
64%
Các thuê bao có kết nối 3G/4G

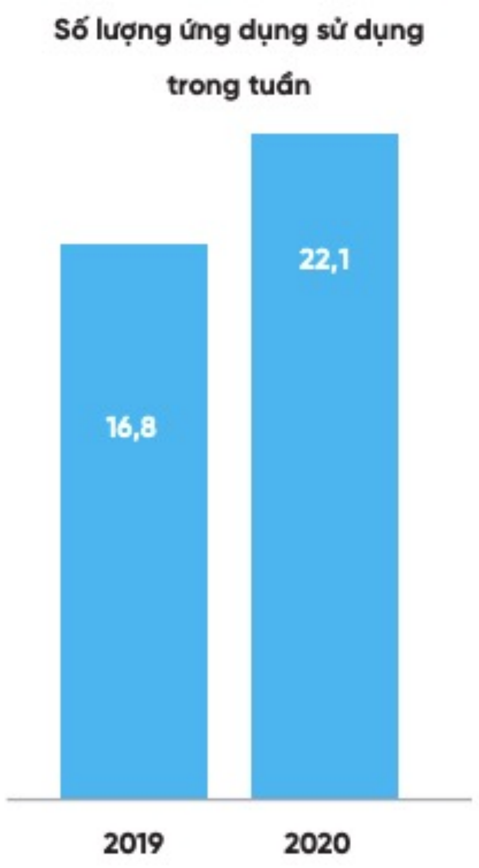
70%
Dân số sử dụng Internet

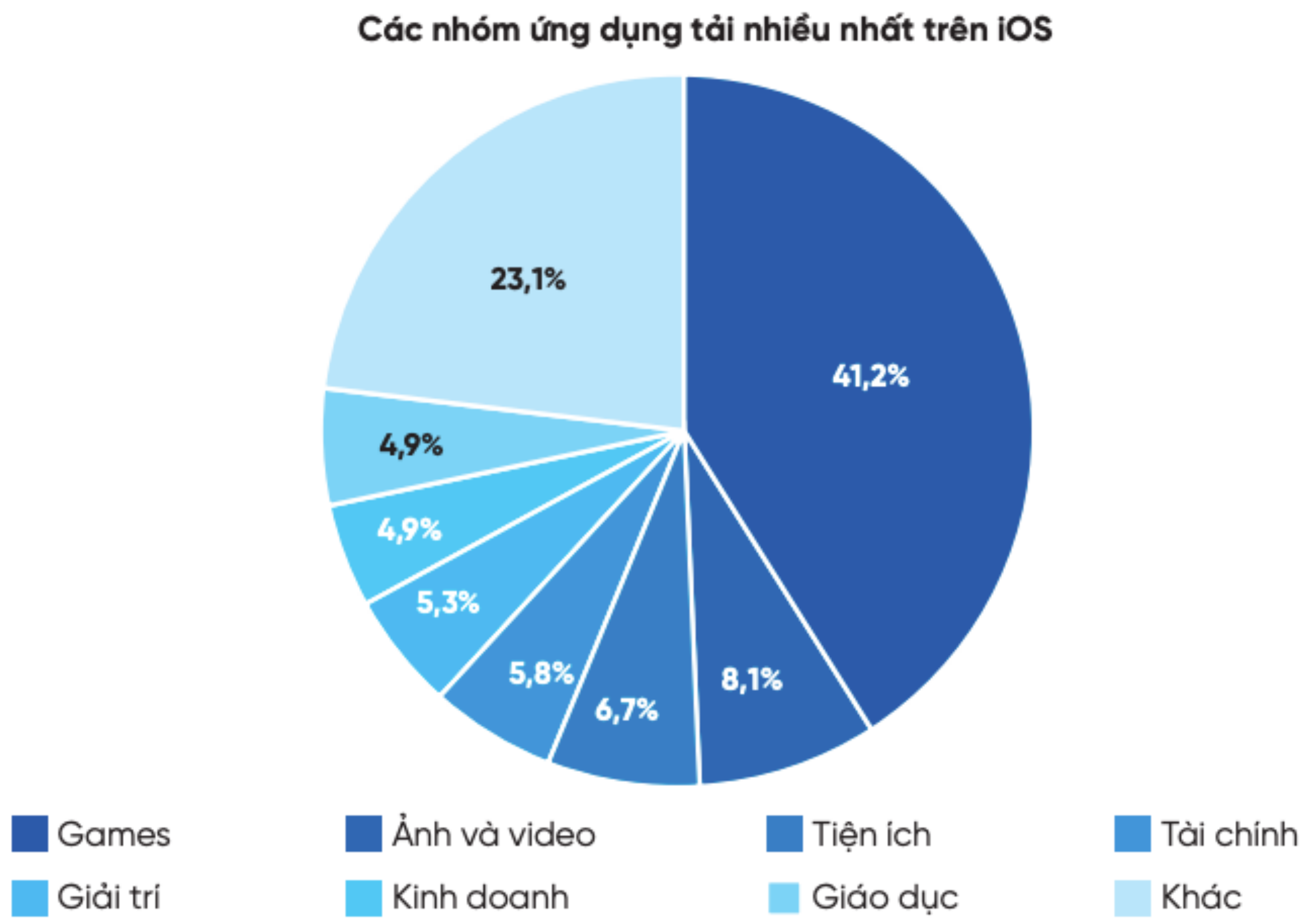
95%
Trong đó sử dụng Internet qua di động

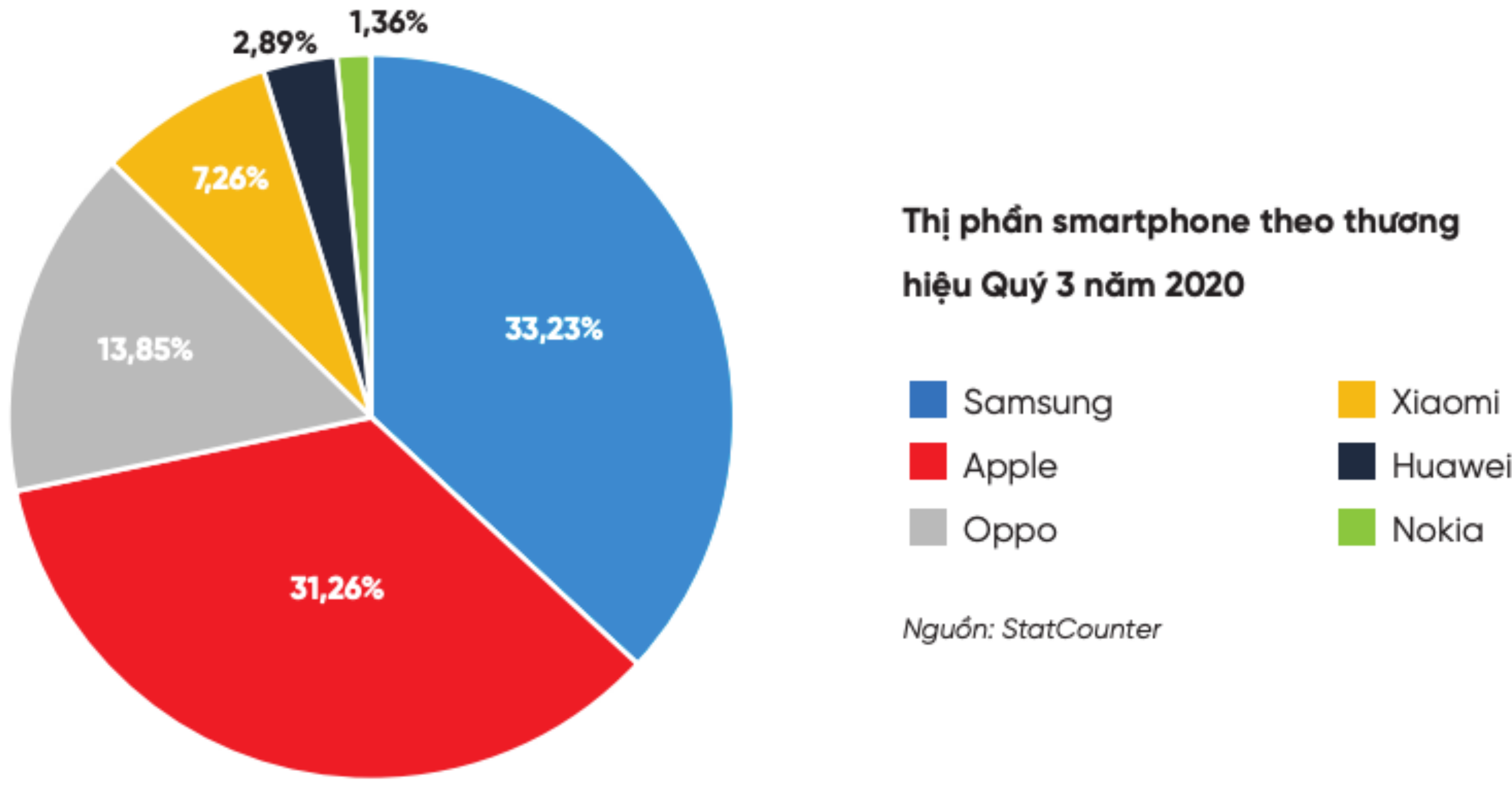
6,5 GIỜ
Thời gian sử dụng Internet trung bình hàng ngày

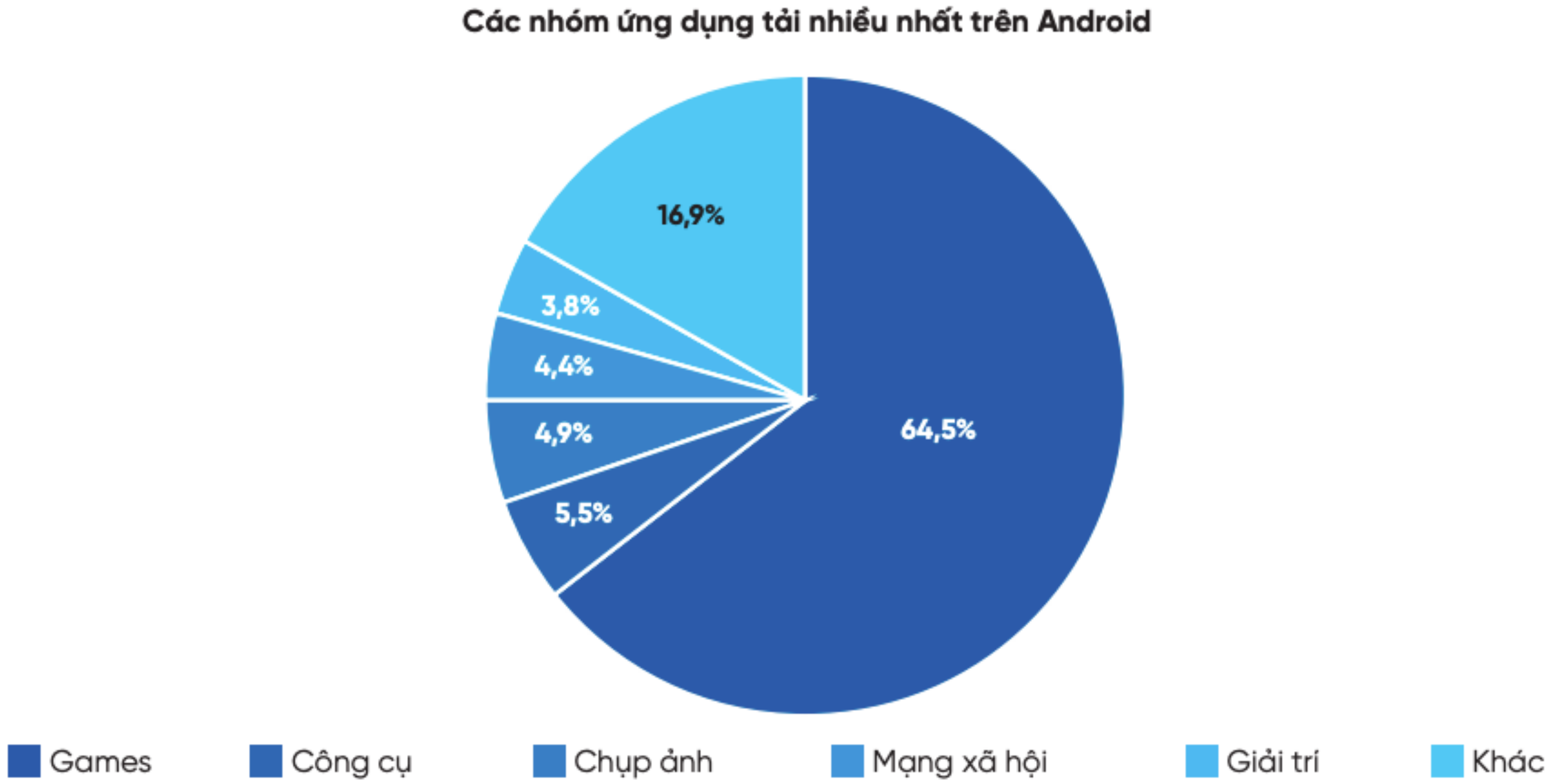
3 GIỜ 18 PHÚT
Thời gian trung bình sử dụng Internet qua di động

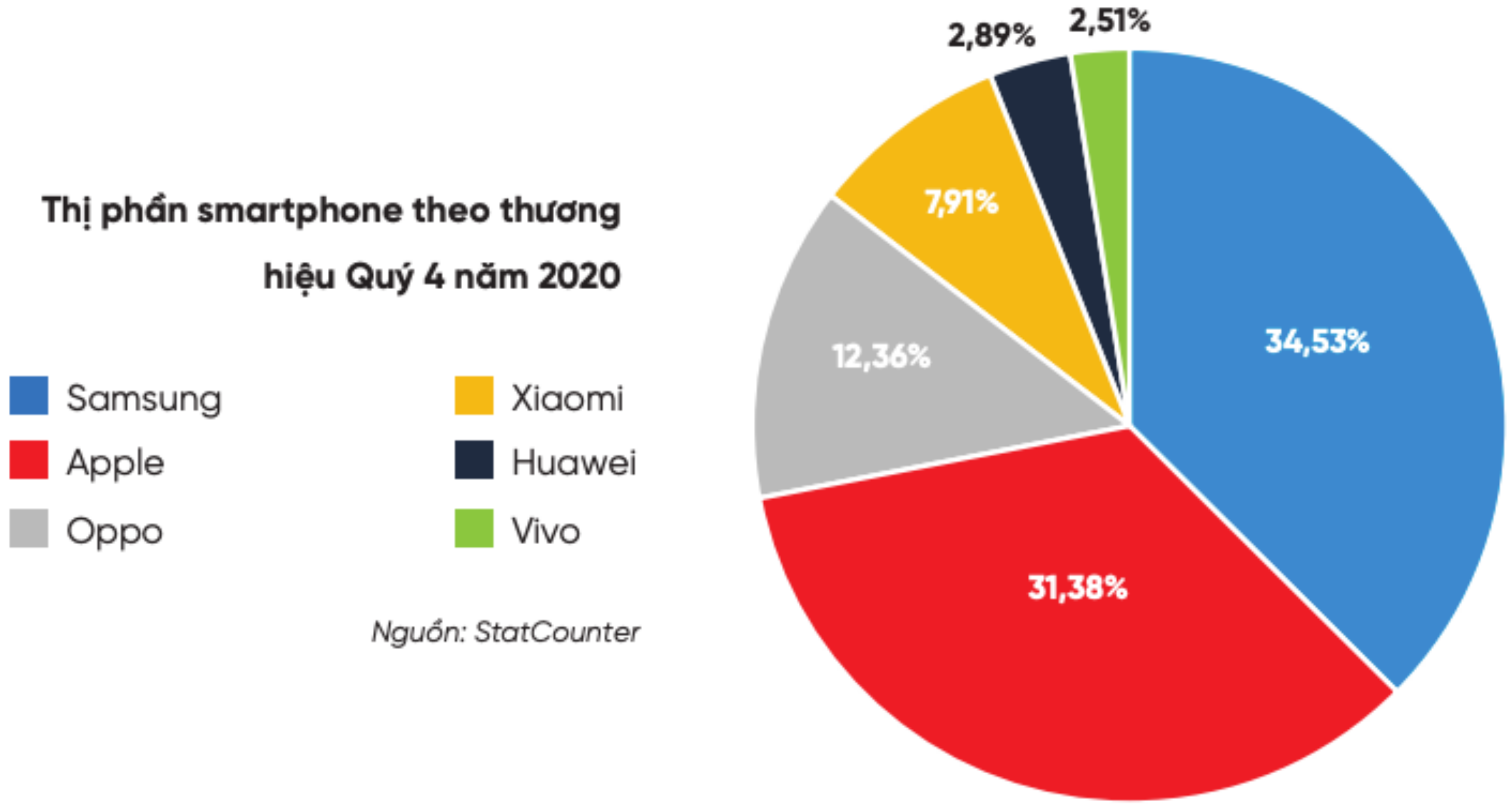




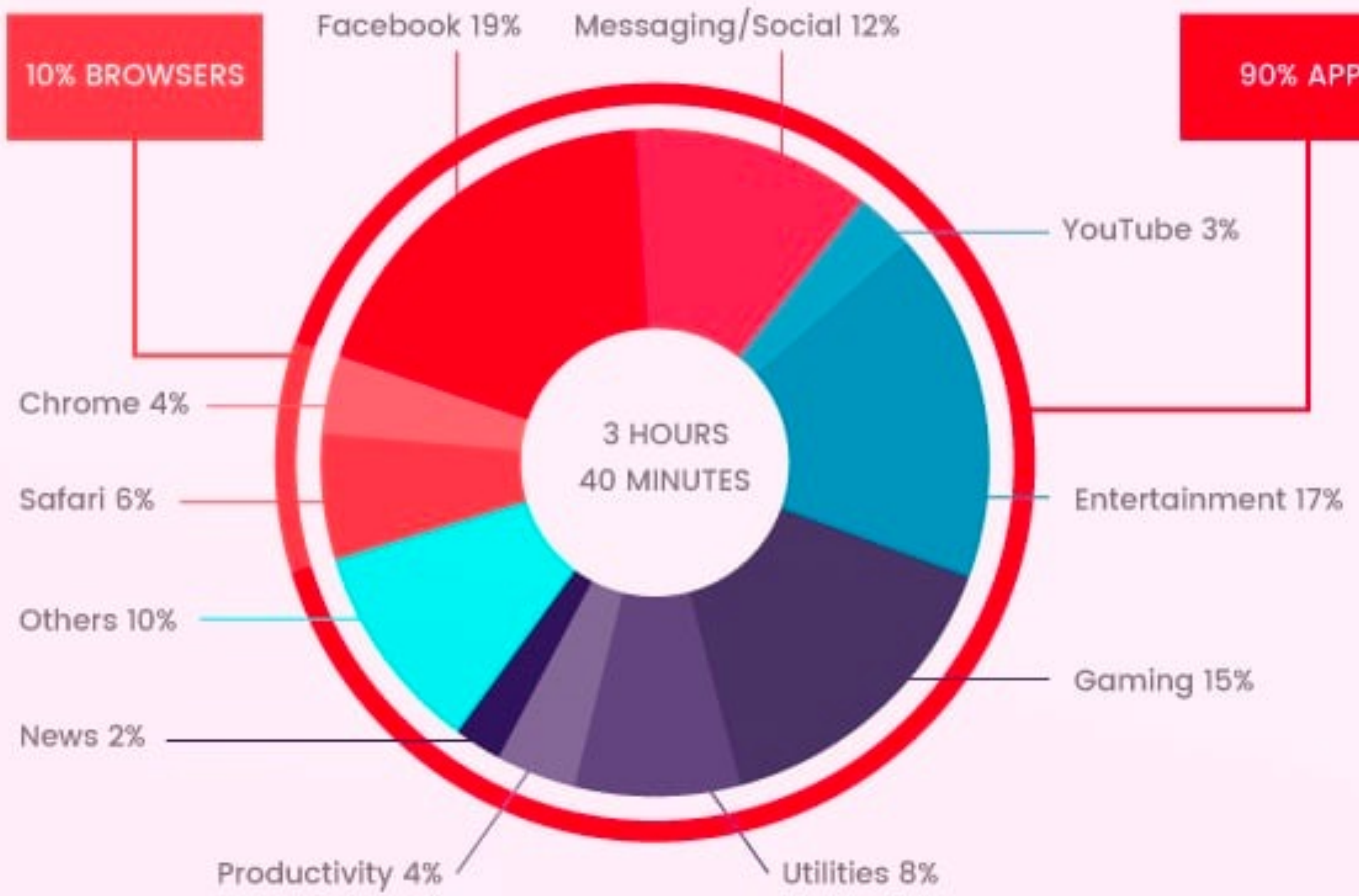

Các nhóm ứng dụng tải nhiều nhất trên iOS
41,2%
8,1%
6,7%
5,8%
5,3%
4,9%
4,9%
23,1%
Games
Ảnh và video
Tiện ích
Tài chính
Giải trí
Kinh doanh
Giáo dục
Khác


33,23%
31,26%
13,85%
7,26%
2,89%
1,36%
Thị phần smartphone theo thương hiệu Quý 3 năm 2020
Samsung
Xiaomi
Apple
Huawei
Oppo
Nokia
Nguồn: StatCounter


Các nhóm ứng dụng tải nhiều nhất trên Android
64,5%
5,5%
4,9%
4,4%
3,8%
16,9%
Games
Công cụ
Chụp ảnh
Mạng xã hội
Giải trí
Khác


Thị phần smartphone theo thương hiệu Quý 4 năm 2020
Samsung
Xiaomi
Apple
Huawei
Oppo
Vivo
Nguồn: StatCounter
34,53%
31,38%
12,36%
7,91%
2,89%
2,51%

90% OF TIME ON MOBILE IS SPENT IN APPS



2. Mobile App Characteristics

What makes mobile phone different from other platforms?

Battery Life
Smartphone have a limited power capacity

Continuous Changes
OS updates and new hardware are introduced every day

Store Distribution
App should be reviewed and distributed via official stores

Unstable Communication
Not always available with 4G/5G Technology, Wifi, BLE, NFC

Small Screen Size
5.9-6.5” screens are the most popular

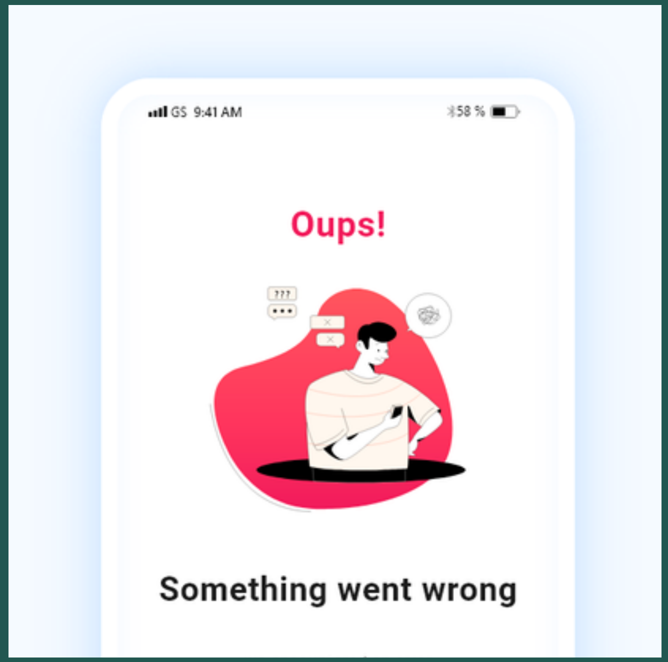
Important Characteristics

Short-Duration Activities

Mobile App activities tend to be short term, ranging from several seconds to several minute

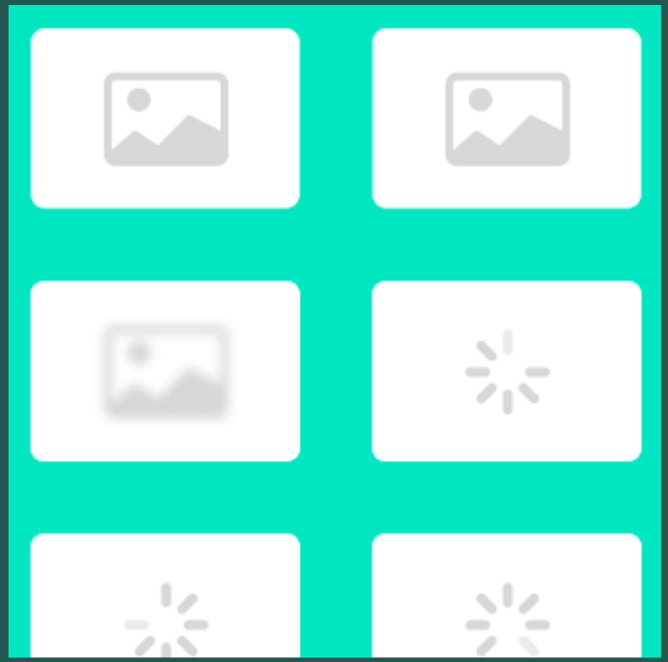
Focused Purpose

Mobile Apps provide services as quickly and as efficiently as possible with little or no navigation.



Responsiveness

Mobile Apps must return some type of acknowledgment immediately upon performing an action on the device.

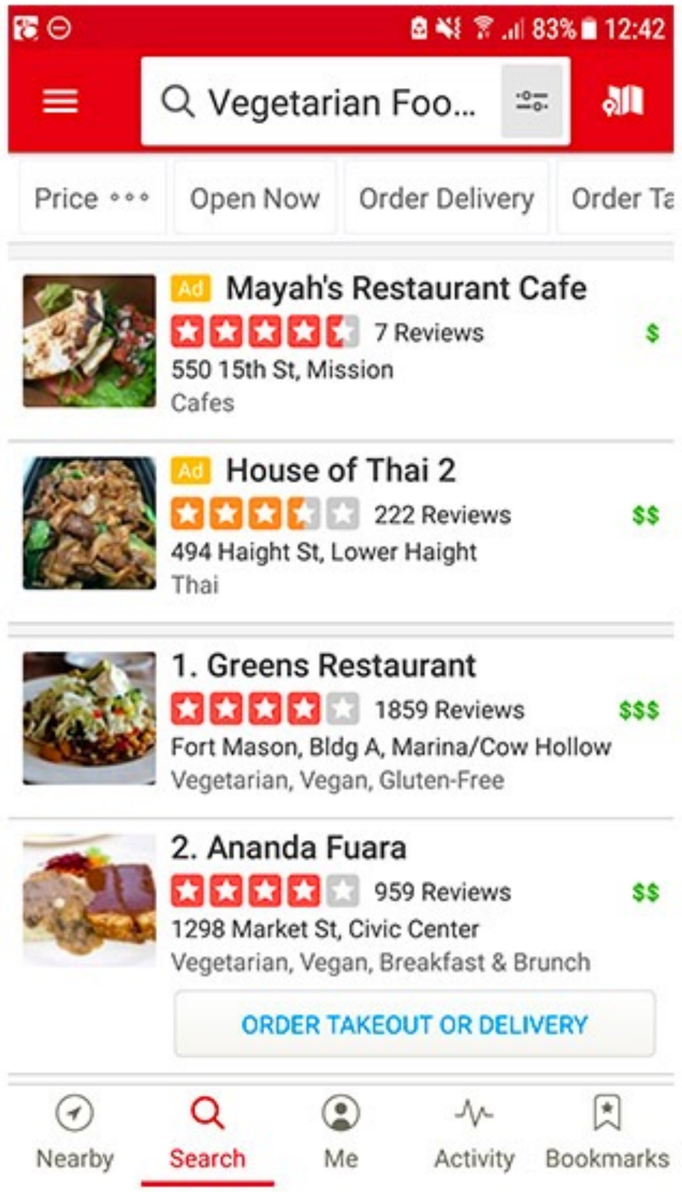


Customized Data Interaction

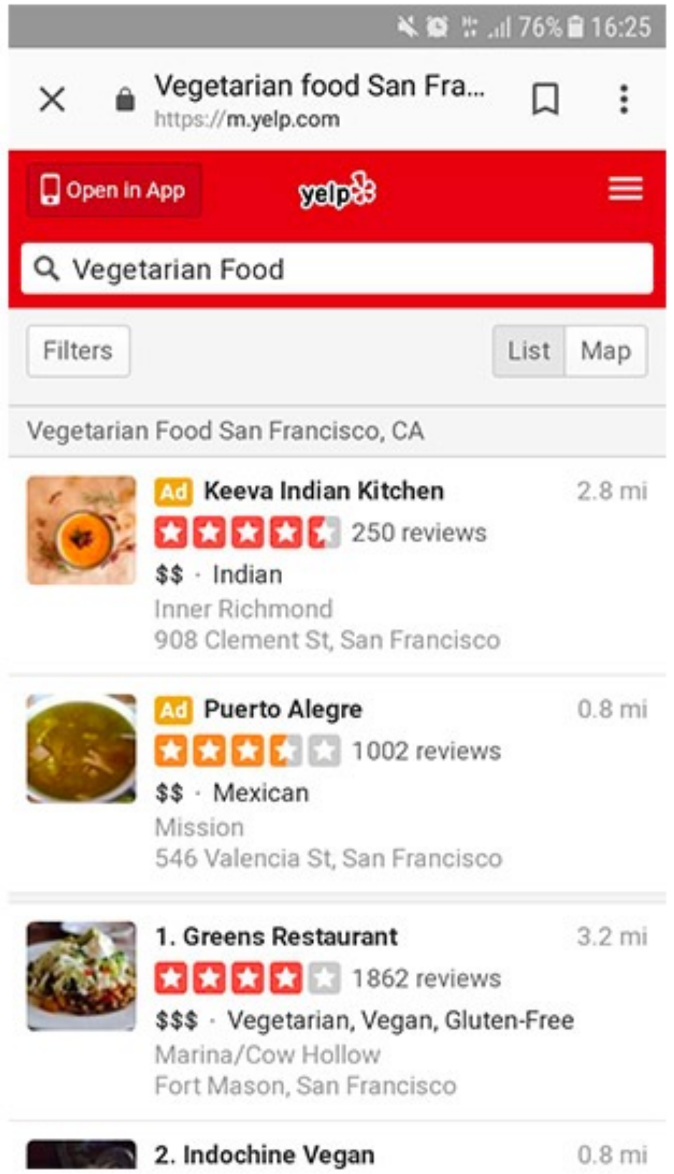
Mobile Apps require limiting the size of the content being downloaded via the Internet

3.Mobile App Platforms

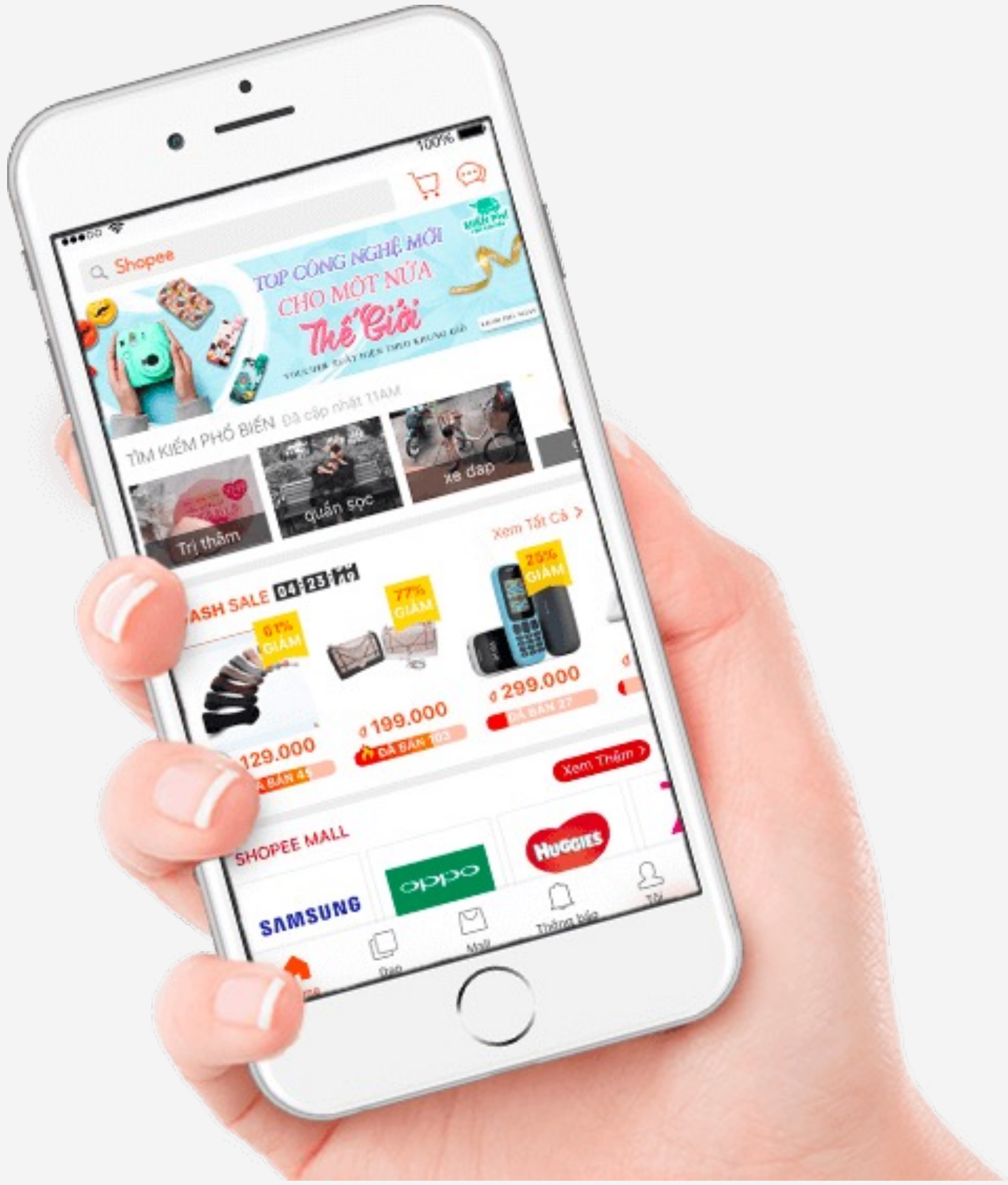
Parameters	Native app development	Cross-platform app development	Hybrid app development	Progressive web app development
Definition	These apps run natively on the platform of your choice	Cross-platform app development is about developing one app and publishing it on both Android and iOS together	It is based on the web view which runs on a web application in a native browser	Progressive web apps are hybrids of regular web pages and native apps
Tools and technologies used	Java, Kotlin, Python, etc.	Xamarin, NativeScript, Node.js, etc.	Objective C, Ionic, HTML5, etc.	Javascript, Ruby, CSS, etc.
Advantages	1. Greater user experience 2. Direct access to the device's APIs	1. Single code for multiple platforms 2. Easy to maintain app	1. Only one codebase is required for a hybrid app 2. Access to device's internal APIs and device hardware	1. The same app runs on web and mobile. Hence, it has less expensive upfronts 2. No need to install or update as it is accessible through URL
Performance	Fastest speed and efficiency as it is directly built on OS's platform	The need for an additional abstraction layer and rendering process makes the cross-platform app slower than its native counterpart	Slower than native and cross-platform apps as they are not usually optimized for their platform	Progressive web apps can be decently fast, but their speed is entirely dependent on internet connection



Mobile App



Web App

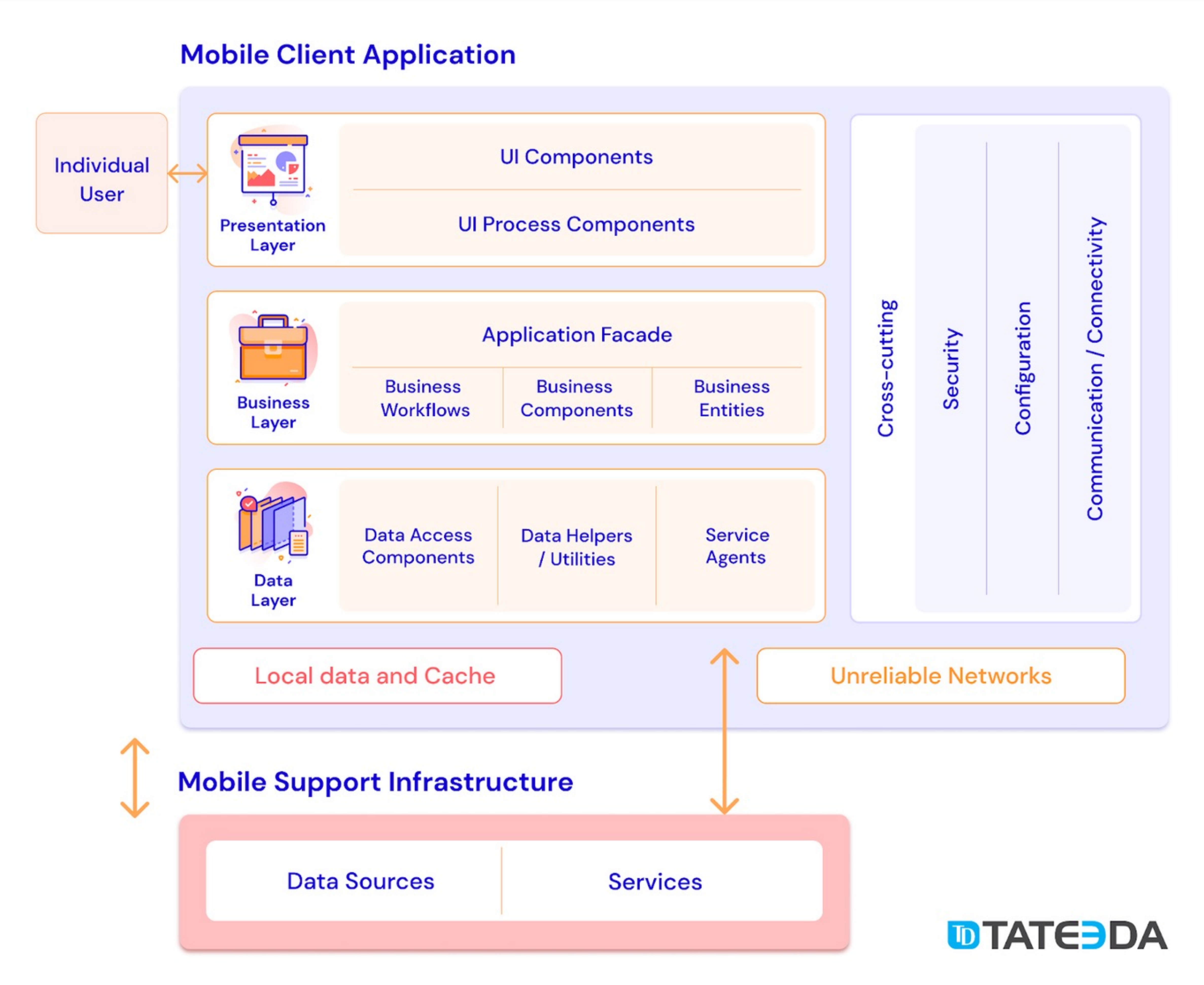


Shopee

	React Native	Ionic	Xamarin	Flutter
	A framework for building native apps with React.	One app running on everything.	Complete Binding for the underlying SDKs.	Makes it easy and fast to build beautiful mobile apps.
	77 401 ★	77 401 ★		64 773 ★
Developers	Facebook	Drifty	Microsoft	Google
Language	JavaScript	TypeScript	C#	Dart
Performance	Close to native	Moderate	Moderate	Amazingcontrolle
GUI	Use native UI controllers	HTML, CSS	Use native UI controllers	Use proprietary widgets and deliver UI
Code reusability	90% of code is reusable	98% of code is reusable	98% of code is reusable	50 - 90% of code is reusable
Testing	Mobile device or emulator	Any browser	Mobile device or emulator, test cloud	Mobile device or emulator
Hot reload	Yes	No	No	Yes
Apps	Airbnb, Discord, Instagram	MarketWatch, Pacifica, JustWatch	Olo, Storyo, APX	KlasterMe, PostMuse Reflectly

4. Mobile App Architecture

The most popular multilayer architecture is the three-layer architecture, together with other supported modules



What is a good Mobile App architecture?

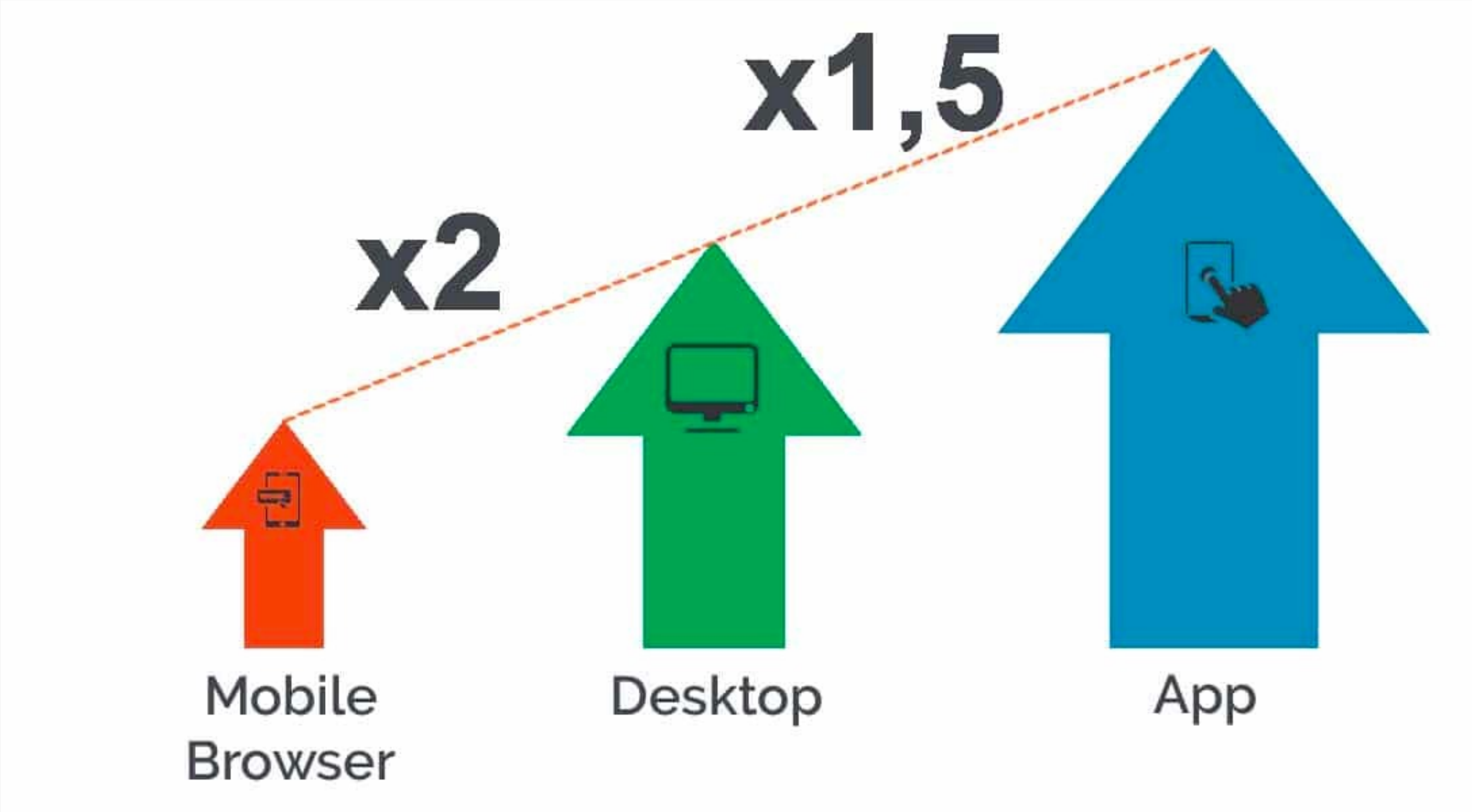
A well-planned mobile app development architecture ensures a number of strengths that will positively affect your long-term development goals and priorities,

- Quality and speed of app performance: if your mobile app is well-debugged and smoothly running, it will obviously be more popular with users than a laggy or frequently-crashing piece of software...
- App compatibility with different types of devices and mobile platforms will enable a wider audience to discover and use your application and this will help you generate more income.
- High levels of scalability and adjustability: is adding new features fast and easy or does it require a massive rethinking of the app structure?
- User-friendliness, usability, and efficiency in executing intended user tasks and functions.
- Killer features and other features that support competitiveness and safeguard good market positions of your app.
- Capabilities for so-called graceful degradation and/or fault-tolerance, which are both necessary to guarantee sustainability and consistent UX of your application.
- Cost-efficient application architecture: is your mobile app designed wisely enough to avoid numerous and costly fixes and revisions in the long run?

https://tateeda.com/blog/fundamentals-of-mobile-application-architecture

5. Mobile App Business

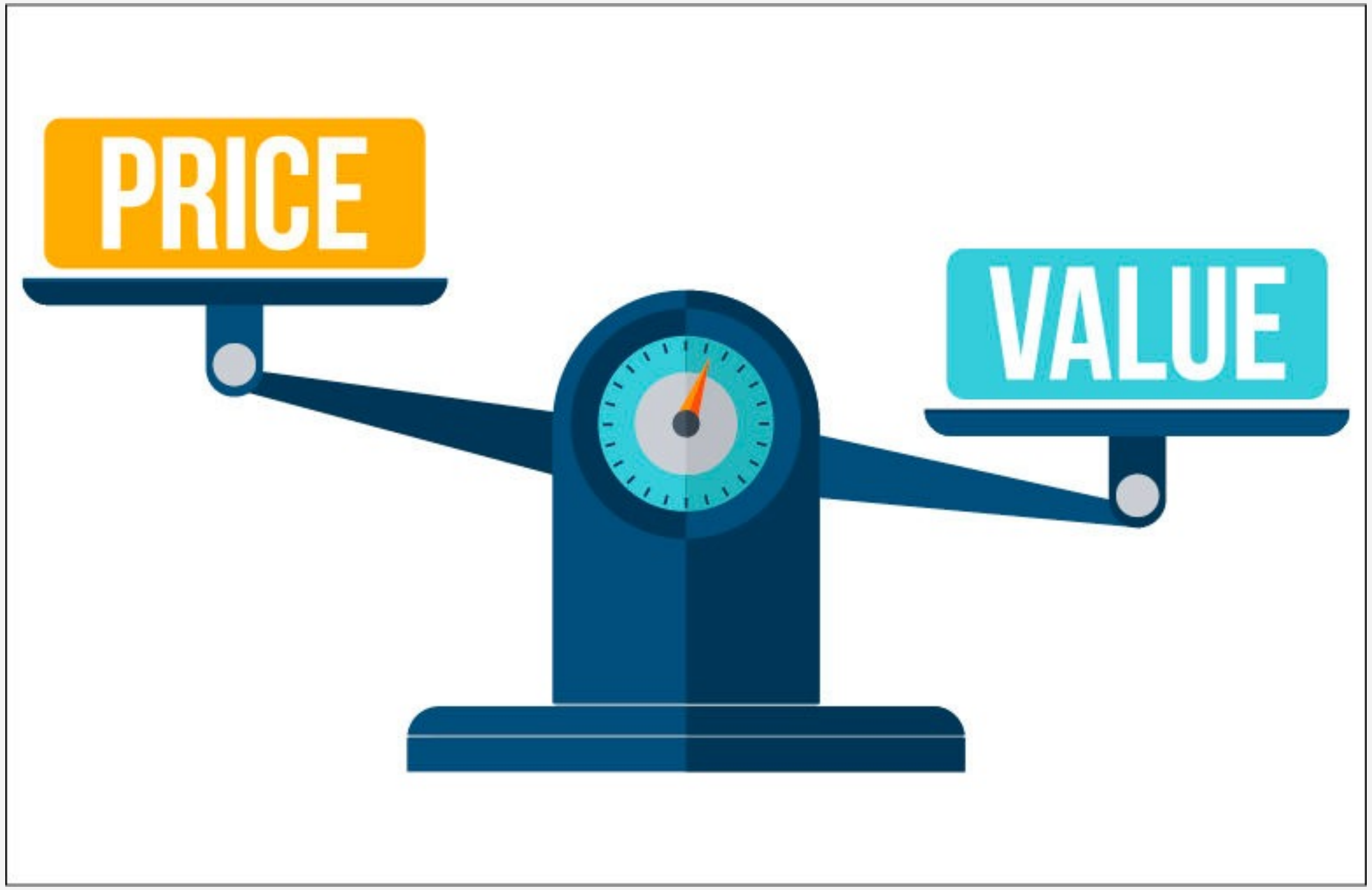
According to Criteo, users view 4.2x more products per session in mobile apps than they do when using mobile websites.



Is a Mobile App always good?

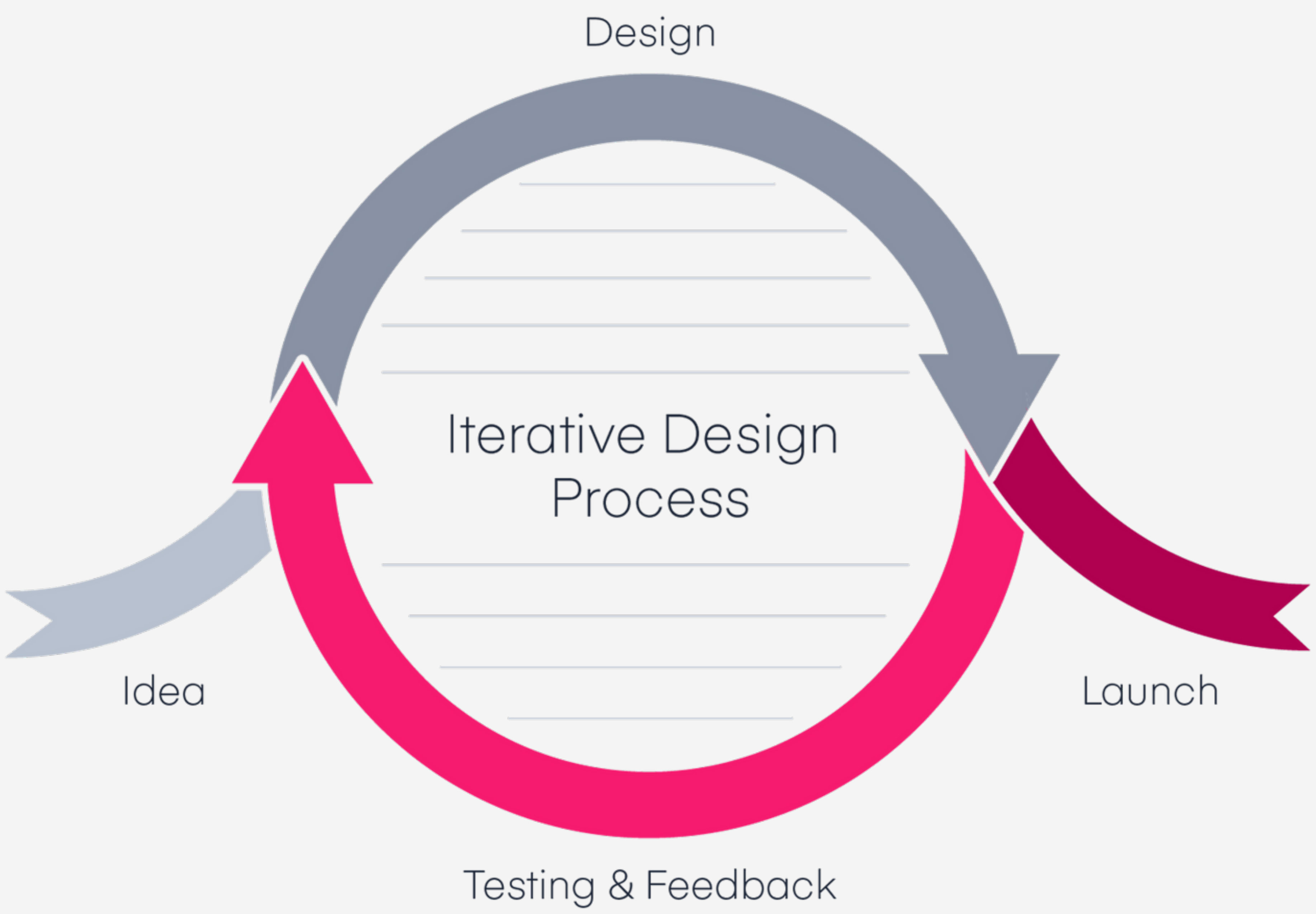
Balance your budget and resources

Don't try to satisfy all your customer's demands



Bing things often have small beginnings

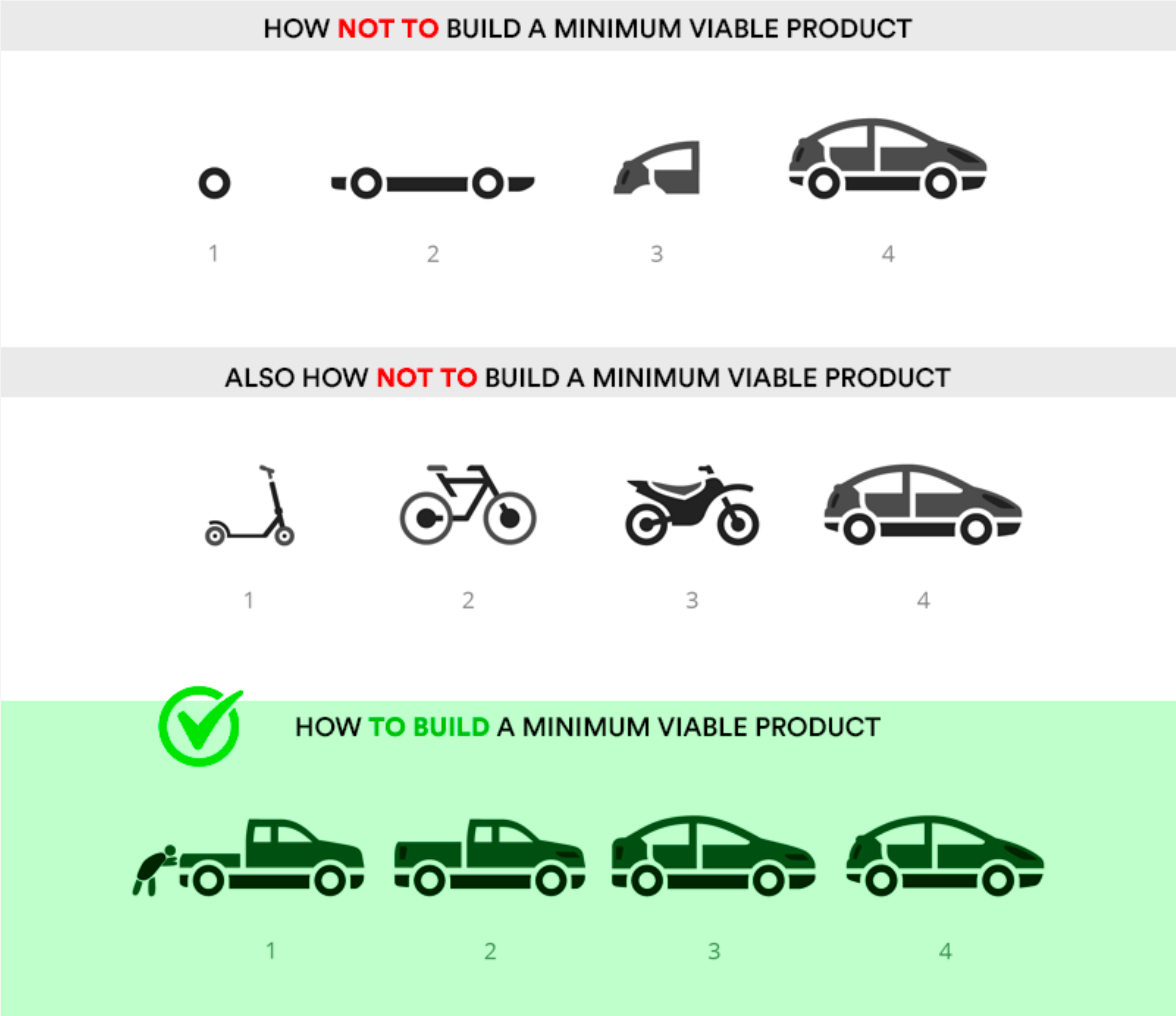
Everything in the world starts small and then becomes bigger—except for bad things. They start big and then get smaller.



Minimum Viable Product (MVP)

MVP is developing a basic version of the app based on must-have features and launching it in the market for customer validation

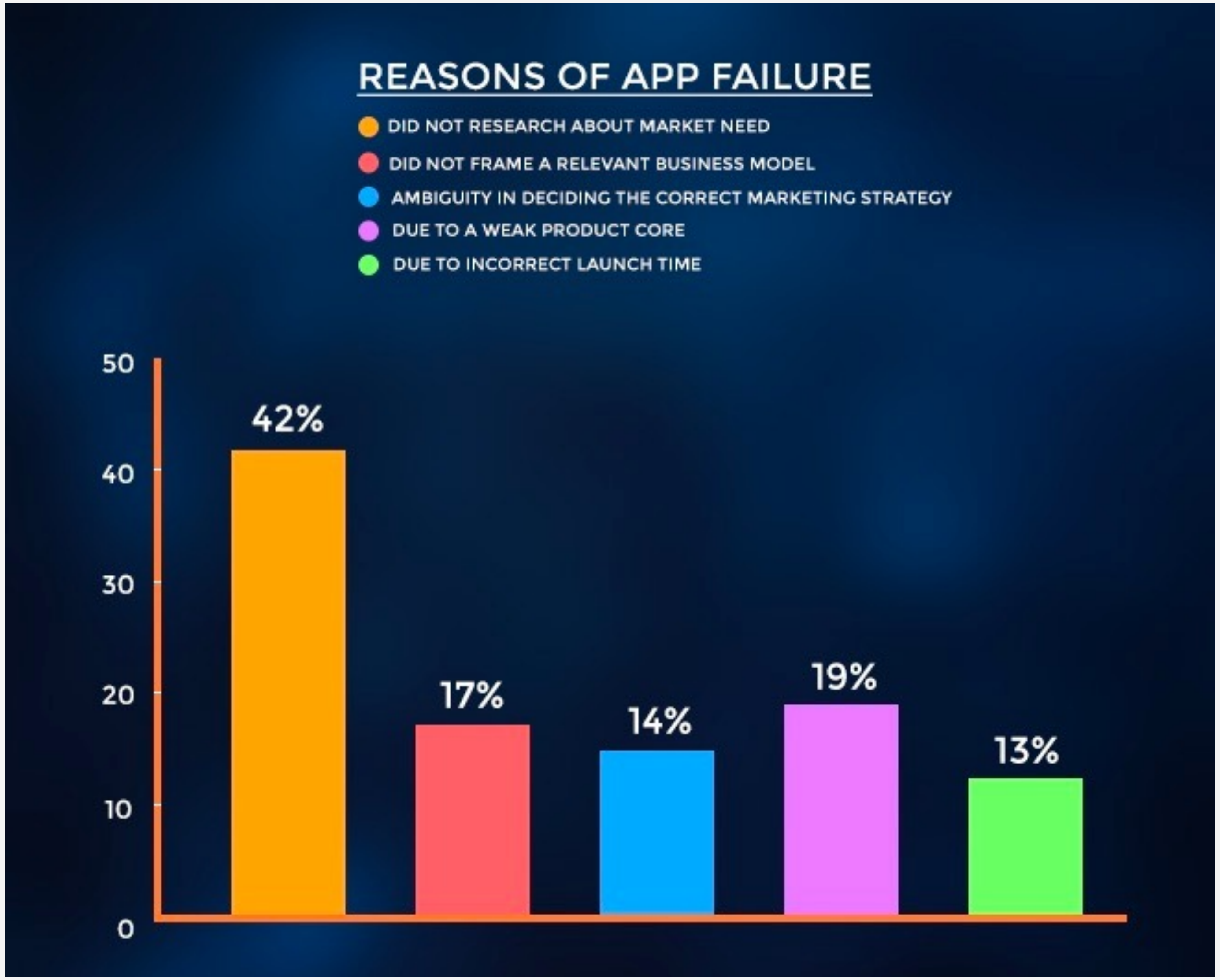
This helps businesses achieve product-market fit while also helping clients save money and time.



Learn from mistakes

The mobile app market is saturated with competition.

Only 20% of downloaded apps see users return after the first use, whereas 3% of apps remain in use after a month.



6.Next Steps

Self-study is always important.

Don't forget to follow these additional steps to empower your career...

...and your score, too.

Explore Mobile App Trends	Study the "Mobile App Report 2021 - APPOTA" and other worldwide reports to catch up with important trends
Learn From Trending Apps	Discover and analyze top-grossing products to improve your app's business.
Understand Business Strategy	Mobile App is not always the most important step to starting a business. Research how to select a suitable platform for any startup project.
Complete Your Weekly Quiz	Quiz or Quit! Don't forget to finish your weekly quiz on time.

Any question?

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